

$$\textcircled{1} \hat{X}(\omega) = -i \cdot \text{sgn}(\omega) X(\omega)$$

$$1) \hat{X}(t) = \frac{1}{\pi t} * X(t) \rightarrow \hat{X}(\omega) = \mathcal{F}\left(\frac{1}{\pi t}\right) \cdot X(\omega)$$

$$2) \mathcal{F}\left(\frac{1}{\pi t}\right) = \int_{-\infty}^{+\infty} \frac{e^{-i\omega t}}{\pi t} dt \Rightarrow \oint \frac{e^{-i\omega z}}{\pi z} dz$$

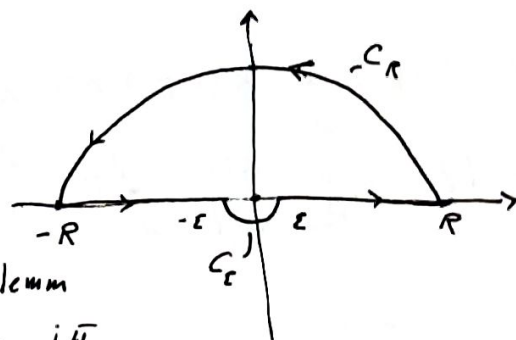
$\omega < 0$:

$$\oint_C \frac{e^{-i\omega z}}{\pi z} dz = 2\pi i \cdot \text{res}_{z=0} \left[\frac{e^{-i\omega z}}{\pi z} \right] = \frac{2\pi i}{\pi}$$

$$\oint_C = \int_{C_R} + \int_{-R}^{-\epsilon} + \int_{C_\epsilon} + \int_{\epsilon}^R ; \int_{C_\epsilon} \frac{e^{-i\omega z}}{\pi z} \xrightarrow{\epsilon \rightarrow 0} \frac{i\pi}{\pi}$$

Jordan's lemma
 $\int_{C_R} \rightarrow 0$
 $R \rightarrow \infty$

$$\oint_C = 2i = 0 + \int_{-R/[-\epsilon; \epsilon]} + i \Rightarrow \int_{-\infty}^{+\infty} \frac{e^{-i\omega t}}{\pi t} dt = i, \omega < 0$$

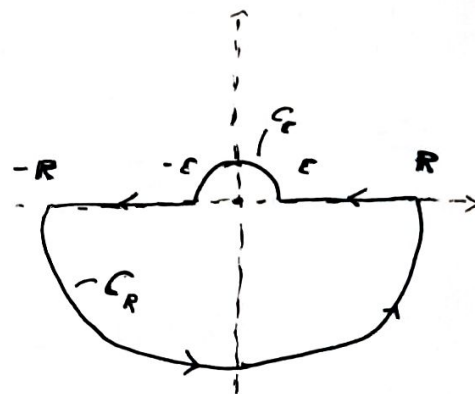


$\omega > 0$:

$$\oint_C \frac{e^{-i\omega z}}{\pi z} dz = 2\pi i \cdot \text{res}_{z=0} \left[\frac{e^{-i\omega z}}{\pi z} \right] = 2i$$

$$\oint_C = \oint_{C_R} + \int_R^\epsilon + \int_{C_\epsilon} + \int_{-\epsilon}^{-R} ; \int_{C_\epsilon} \frac{e^{-i\omega z}}{\pi z} \xrightarrow{\epsilon \rightarrow 0} i$$

$$\int_{C_R} \rightarrow 0 \quad R \rightarrow \infty ; \oint_C = 2i \neq 0 + \int_{-R/[-\epsilon; \epsilon]} + i \Rightarrow \int_{-\infty}^{+\infty} \frac{e^{-i\omega t}}{\pi t} dt = -i$$



$$\mathcal{F}\left(\frac{1}{\pi t}\right) = \begin{cases} i, & \omega < 0 \\ -i, & \omega \geq 0 \end{cases} = -i \text{sgn}(\omega)$$

$$\text{Result: } \hat{X}(\omega) = (-i \text{sgn}(\omega)) \cdot X(\omega)$$

②

a) $x(t) = x(-t) \Rightarrow \hat{x}(t) = -\hat{x}(-t)$

$$\begin{aligned} \hat{x}(t) &= \int_{-\infty}^{+\infty} x(\alpha) \cdot \frac{1}{\pi(t-\alpha)} d\alpha \stackrel{\alpha=\mu}{=} - \int_{+\infty}^{-\infty} \frac{x(-\mu) d\mu}{\pi(t+\mu)} \stackrel{x(-\mu)=x(\mu)}{=} \int_{-\infty}^{+\infty} \frac{x(\mu) d\mu}{(-1)\pi(t-\mu)} \\ &= - \int_{-\infty}^{+\infty} \frac{x(\mu) d\mu}{\pi(t-\mu)} = -\hat{x}(-t) \end{aligned}$$

b) $x(t) = -x(-t) \Rightarrow \hat{x}(t) = \hat{x}(-t)$

$$\begin{aligned} \hat{x}(t) &= \int_{-\infty}^{+\infty} \frac{x(\alpha) d\alpha}{\pi(t-\alpha)} \stackrel{\alpha=\mu}{=} - \int_{+\infty}^{-\infty} \frac{x(-\mu) d\mu}{\pi(t+\mu)} = \left\{ \begin{array}{l} x(t) = -x(-t) \Rightarrow \\ \Rightarrow x(-t) = -x(t) \end{array} \right\} \\ &= \int_{-\infty}^{+\infty} \frac{-x(\mu) d\mu}{\pi(t+\mu)} = \int_{-\infty}^{+\infty} \frac{x(\mu) d\mu}{\pi(t-\mu)} = \hat{x}(-t) \end{aligned}$$

c) $\hat{\hat{x}}(t) = -x(t)$

$$\hat{x}(\omega) = -i \operatorname{sgn}(\omega) X(\omega) = \mathcal{F}[\hat{x}(t)]$$

$$\hat{x}(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} -i \operatorname{sgn}(\omega) X(\omega) e^{i\omega t} d\omega$$

$$\begin{aligned} \hat{\hat{x}}(t) &= \mathcal{F}^{-1}[-i \operatorname{sgn}(\omega) \cdot \mathcal{F}[\hat{x}(t)]] = \mathcal{F}^{-1}[-i \operatorname{sgn}(\omega) \cdot (-i \operatorname{sgn}(\omega) X(\omega))] \\ &= \mathcal{F}^{-1}[(-1) \operatorname{sgn}^2(\omega) X(\omega)] = \\ &= \mathcal{F}^{-1}[(-1) X(\omega)] = -x(t) \end{aligned}$$

$$\operatorname{sgn}^2(\omega) = \begin{cases} 0, & \omega = 0 \\ 1, & \omega \neq 0 \end{cases}$$

d) $\int_{-\infty}^{+\infty} x^2(t) dt = \int_{-\infty}^{+\infty} \hat{x}^2(t) dt$

$$\int_{-\infty}^{+\infty} x^2(t) dt = \int_{-\infty}^{+\infty} \frac{1}{4\pi^2} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} X(\omega) X(\tilde{\omega}) e^{i\omega t} e^{i\tilde{\omega} t} d\omega d\tilde{\omega} dt =$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} X(\omega) X(\tilde{\omega}) d\omega d\tilde{\omega} \cdot \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{i(\omega+\tilde{\omega})t} dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} X(\omega) X(\tilde{\omega}) \cdot$$

$$\cdot \delta(\tilde{\omega} - (-\omega)) d\omega d\tilde{\omega} = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\omega) X(-\omega) d\omega = I_1$$

$$\int_{-\infty}^{+\infty} \hat{x}(t) dt = \int_{-\infty}^{+\infty} \frac{1}{4\pi^2} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} -i \operatorname{sgn}(\omega) X(\omega) (-i) \operatorname{sgn}(\tilde{\omega}) X(\tilde{\omega}) e^{i(\omega+\tilde{\omega})t} d\omega d\tilde{\omega} dt =$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \delta(\tilde{\omega} - (-\omega)) (-1) \operatorname{sgn}(\omega) \operatorname{sgn}(\tilde{\omega}) X(\omega) X(\tilde{\omega}) d\omega d\tilde{\omega} = \frac{1}{2\pi} \int_{-\infty}^{+\infty} (-1) \cdot$$

$$\operatorname{sgn}(\omega) \operatorname{sgn}(-\omega) X(\omega) X(-\omega) d\omega = \begin{cases} \operatorname{sgn}(\omega) \cdot \operatorname{sgn}(-\omega) = -1 \\ \end{cases}$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\omega) X(-\omega) d\omega = I_2$$

$$I_1 = I_2$$

e) $\int_{-\infty}^{+\infty} x(t) \hat{x}(t) dt = 0$

$$\int_{-\infty}^{+\infty} x(t) \hat{x}(t) dt = \int_{-\infty}^{+\infty} \frac{1}{4\pi^2} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x(\tilde{\omega}) (-i) \operatorname{sgn}(\omega) X(\omega) e^{i(\omega+\tilde{\omega})t} d\omega d\tilde{\omega} dt =$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \delta(\tilde{\omega} - (-\omega)) X(\tilde{\omega}) (-i) \operatorname{sgn}(\omega) X(\omega) d\omega d\tilde{\omega} = \frac{-i}{2\pi} \int_{-\infty}^{+\infty} X(-\omega) X(\omega) \cdot$$

$$\cdot \operatorname{sgn}(\omega) d\omega = \begin{cases} f(\omega) = X(-\omega) X(\omega) \operatorname{sgn}(\omega) \\ f(\omega) = -f(-\omega) \\ X(-\omega) X(\omega) \operatorname{sgn}(\omega) = -X(\omega) X(-\omega) \operatorname{sgn}(-\omega) \end{cases} = \frac{-i}{2\pi} \int_0^{+\infty} f(\omega) d\omega +$$

$$+ \frac{(-i)}{2\pi} \int_{-\infty}^0 f(\omega) d\omega = \frac{-i}{2\pi} \int_0^{+\infty} f(\omega) d\omega + \int_0^{+\infty} f(-\omega) d\omega = \frac{-i}{2\pi} \int_0^{+\infty} (f(\omega) - f(\omega)) d\omega = 0$$

$$d\omega = 0$$

③ $\hat{X}(t) = ?$

a) $x(t) = \text{const} = 1$

$$\hat{X}(t) = \int_{-\infty}^{+\infty} \frac{1}{\pi \mu} d\mu = \frac{1}{\pi} \ln(|\mu|) \Big|_{-\infty}^{+\infty} = 0$$

b) $x(t) = \sin(\omega t)$; $\mathcal{F}[\sin(\omega t)] = \frac{\pi}{i} (\delta(\omega + \omega) + \delta(\omega - \omega))$

$$\begin{aligned} \hat{X}(t) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} (-i) \operatorname{sgn}(\omega) \left(\frac{\pi}{i}\right) (-\delta(\omega + \omega) + \delta(\omega - \omega)) e^{i\omega t} d\omega = \\ &= \left(-\frac{1}{2}\right) \int_{-\infty}^{+\infty} \underbrace{\operatorname{sgn}(-\omega)}_{-\operatorname{sgn}(\omega)} e^{-i\omega t} + \operatorname{sgn}(\omega) e^{i\omega t} d\omega = \frac{-\operatorname{sgn}(\omega) \cos(\omega t)}{-\operatorname{sgn}(\omega) \cos(\omega t)} \end{aligned}$$

c) $x(t) = \cos(\omega t)$; $\mathcal{F}(\cos(\omega t)) = \pi (\delta(\omega - \omega) + \delta(\omega + \omega))$

$$\begin{aligned} \hat{X}(t) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} (-i) \operatorname{sgn}(\omega) \pi (\delta(\omega - \omega) + \delta(\omega + \omega)) e^{i\omega t} d\omega = \\ &= \left(\frac{-i}{2}\right) \left(\operatorname{sgn}(\omega) e^{i\omega t} + \underbrace{\operatorname{sgn}(-\omega)}_{-\operatorname{sgn}(\omega)} e^{-i\omega t} \right) = \left(\frac{-i \operatorname{sgn}(\omega)}{2}\right) (e^{i\omega t} - e^{-i\omega t}) = \\ &= \operatorname{sgn}(\omega) \sin(\omega t) \end{aligned}$$

d) $x(t) = \frac{1}{1+t^2}$

1) $\mathcal{F}[x(t)] = \int_{-\infty}^{+\infty} \frac{e^{-i\omega t}}{(t+i)(t-i)} dt = \int_{-\infty}^{+\infty} \frac{e^{-i\omega z}}{(z+i)(z-i)} dz = I$

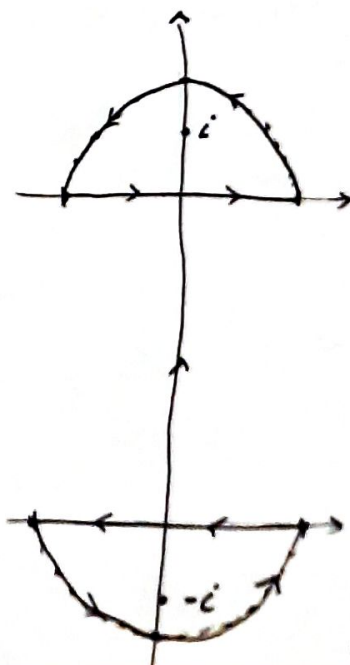
$\omega < 0$:

$$I = \oint_C \frac{e^{-i\omega z}}{(z+i)(z-i)} dz = 2\pi i \operatorname{res}_{z=i} \left[\frac{e^{-i\omega z}}{(z+i)(z-i)} \right] = 2\pi i \frac{e^{-\omega}}{2i} = \pi e^{-\omega}$$

$\omega > 0$:

$$I = -\oint_C \frac{e^{-i\omega z}}{(z+i)(z-i)} dz = -2\pi i \operatorname{res}_{z=-i} \left[\frac{e^{-i\omega z}}{(z+i)(z-i)} \right] = -2\pi i \frac{e^{-\omega}}{-2i} = \pi e^{-\omega}$$

$$\mathcal{F}[x(t)] = \pi e^{-|\omega|}$$



$$\begin{aligned}
 2) \hat{X}(t) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} (-i) \operatorname{sgn}(\omega) \pi e^{-i|\omega|} e^{i\omega t} d\omega = \frac{-i\pi}{2\pi} \int_0^{+\infty} \operatorname{sgn}(\omega) e^{-\omega} e^{i\omega t} d\omega \\
 &+ \frac{(-i)}{2} \int_{-\infty}^0 \operatorname{sgn}(\omega) e^{\omega} e^{i\omega t} d\omega = \left(\frac{-i}{2}\right) \left[\int_0^{+\infty} e^{-\omega(1-it)} d\omega - \int_{-\infty}^0 e^{\omega(1+it)} d\omega \right] = \\
 &= \left(\frac{-i}{2}\right) \cdot \left[\frac{e^{-\omega(1-it)}}{(it-1)} \Big|_0^{+\infty} - \frac{e^{\omega(1+it)}}{(1+it)} \Big|_{-\infty}^0 \right] = \left(\frac{-i}{2}\right) \cdot \left[\frac{(-1)}{(it-1)} - \frac{1}{(1+it)} \right] = \\
 &= \frac{-i}{2} \cdot \left[\frac{-1-it - it+1}{(-t^2-1)} \right] = \frac{-i}{2} \cdot \frac{-2it}{(-t^2-1)} = \frac{t}{t^2+1}
 \end{aligned}$$

(4)

$$a) e_1(t) = \begin{pmatrix} 1(t) \\ 0; 1 \end{pmatrix}; e_2(t) = \begin{pmatrix} 1(t) \\ 1; 2 \end{pmatrix}; e_3(t) = \begin{pmatrix} 1(t) \\ 2; 3 \end{pmatrix}$$

Dimension of basis = 3

$$b) S_1(t) = (2, 2, 2); S_2(t) = (2, 0, 0, 0)$$

$$S_3(t) = (0, -2, -2, 0)$$

$$S_1(t) = (2, 2, 2); S_2(t) = (2, 0, 0)$$

$$S_3(t) = (0, -2, -2); S_4(t) = (2, 2, 0)$$

$$c) d(S_i, S_j) = \sqrt{\langle S_i - S_j, S_i - S_j \rangle}; i, j = 1, \dots, 4; i \neq j$$

& $i \neq j$:

$$d(S_1, S_2) = 2\sqrt{2}$$

$$d(S_1, S_3) = 6$$

$$d(S_1, S_4) = 2$$

$$d(S_2, S_3) = 2\sqrt{3}$$

$$\Rightarrow d_{\min} = 2$$

$$d(S_2, S_4) = 2$$

$$d(S_3, S_4) = 2\sqrt{6}$$

⑤ $Z(t) = X(t) + i Y(t)$; $S_x(f) = S_y(f) = \begin{cases} N_0, & |f| \leq W \\ 0, & |f| > W \end{cases}$;
 $X(t), Y(t) \in \mathbb{R}$; $E[X(t)] = E[Y(t)] = 0$; $\text{cov}(X, Y) = 0$;
 $X(t), Y(t)$ - stationary random process $X \& Y$ independent

a) $R_x(z) = \frac{1}{2\pi} \int_{-W}^W N_0 e^{i2\pi f z} df = \frac{N_0 W}{\pi} \text{sinc}(2\pi W z)$

$R_y(z) = \{\text{analog}\} = \frac{N_0 W}{\pi} \text{sinc}(2\pi W z)$

$E[Z(t)] = E[X(t) + i Y(t)] \stackrel{X \& Y \text{ indep.}}{=} E[X(t)] + i E[Y(t)] = 0$

$R_z(t+z, t) = E[Z(t+z) \cdot Z^*(t)] = E[(X(t+z) + i Y(t+z)) \cdot (X(t) - i Y(t))] =$
 $= E[(X(t+z) + i Y(t+z)) \cdot (X(t) - i Y(t))] = E[X(t+z) X(t)] +$
 $+ E[Y(t+z) Y(t)] + i E[Y(t+z) X(t)] - i E[X(t+z) Y(t)] = R_x(z) +$
 $+ R_y(z) = \frac{2N_0 W}{\pi} \text{sinc}(2\pi W z)$

$R_z(t+z, t) = R_z(z) \Rightarrow$
 $E[Z(t)] = \text{const}$
 $D[Z(t)] = R_z(0) = \text{const} \Rightarrow Z(t) - \text{stationary process}$

b) $S_z(f) = \int_{-\infty}^{+\infty} R_z(z) e^{-2\pi i f z} dz = \frac{2N_0 W}{\pi} \int_{-\infty}^{+\infty} \text{sinc}(2\pi W z) e^{-2\pi i f z} dz =$
 $= 2N_0 \cdot \mathbb{1}_{[-W, W]}(f)$

c) $\{\varphi_i(t)\}_{i=1}^n$; $Z_j = \int_{-\infty}^{+\infty} Z(t) \varphi_j^*(t) dt$; $\Phi_j(f) = \int_{-\infty}^{+\infty} \varphi_j(t) e^{-2\pi i f t} dt$ $t \in [-W, W]$
 φ_j
 $\langle \varphi_j, \varphi_k \rangle = \delta_{j,k}$
 $E[Z_j] = ?$, $E[Z_j Z_k^*] = ?$, $\{Z_j\}_{j=1}^n \xrightarrow{?} \text{i.i.d.}$, $D[Z_j] = ?$

$$E[Z_j] = E\left[\int_{-\infty}^{+\infty} Z(t) \varphi_j^*(t) dt\right] = \int_{-\infty}^{+\infty} E[Z(t)] \varphi_j^*(t) dt = 0$$

$$\begin{aligned} E[Z_j Z_k^*] &= E\left[\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (Z(t) \varphi_j^*(t)) (Z(z) \varphi_k(z)) dt dz\right] = \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} E[Z(t) Z(z)] \varphi_j^*(t) \varphi_k(z) dt dz = \left\{ \begin{array}{l} \text{WLOG: } t > z \Rightarrow \\ \Rightarrow t - z = \tau \end{array} \right\}^2 \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} R_z(\tau) \varphi_j^*(\tau + z) \varphi_k(z) d\tau dz = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} R_z(t-z) \varphi_j^*(t) \varphi_k(z) dt dz = \\ &= \int_{-\infty}^{+\infty} \varphi_j^*(t) \left(\int_{-\infty}^{+\infty} \varphi_k(z) R_z(t-z) dz \right) dt = \int_{-\infty}^{+\infty} \varphi_j^*(t) (\varphi_k * R_z)(t) dt \quad \textcircled{=} \end{aligned}$$

$$\begin{aligned} \mathcal{F}(\varphi_k * R_z)(t) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \underbrace{\int_{-\infty}^{+\infty} \varphi_k(f) \cdot \varphi_k(f) e^{i2\pi f t} df}_{= 2N_0 \cdot \Pi(f)} = \frac{2N_0}{2\pi} \int_{-W}^W \varphi_k(f) e^{i2\pi f t} df = \\ &= \left\{ \begin{array}{l} \varphi_k(f), f \in [-W, W] \Rightarrow \\ \Rightarrow \varphi_k(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \varphi_k(f) e^{i2\pi f t} df \end{array} \right\} = 2N_0 \cdot \varphi_k(t) \end{aligned}$$

$$\begin{aligned} \textcircled{=} \int_{-\infty}^{+\infty} \varphi_j^*(t) \varphi_k(t) \cdot 2N_0 dt &= 2N_0 \delta_{jk} \Rightarrow D[Z_j] = E[Z_j Z_j^*] - \\ &- |E[Z_j]|^2 = 2N_0 \end{aligned}$$

$$\begin{aligned} Z(t_{\text{fix}}) &= \underbrace{X(t_{\text{fix}})}_{\text{Gaus}} + i \underbrace{Y(t_{\text{fix}})}_{\text{Gaus}} \rightarrow \text{Gaussian} \\ \int_{-\infty}^{+\infty} Z(t) \varphi_j^*(t) dt &= \lim_{d(z) \rightarrow 0} \sum_{k=1}^{n(z)} Z(t_k) \varphi_j^*(t_k) \Delta t_k = \underbrace{\sum_{k=1}^{n(z)} \underbrace{\varphi_j^*(t_k) \Delta t_k}_{\Delta_k} Z(t_k)}_{\text{Gaus}} \\ &= \lim_{d(z) \rightarrow 0} \left(\sum_{k=1}^{n(z)} \underbrace{(\varphi_j^*(t_k) \Delta t_k)}_{\Delta_k} Z(t_k) \right) = \lim_{d(z) \rightarrow 0} \underbrace{\left(\sum_{k=1}^{n(z)} \Delta_k \underbrace{Z(t_k)}_{\text{Gaus}} \right)}_{\text{Gaus}} \Rightarrow \\ &\Rightarrow Z_j - \text{Gaussian random variable} \end{aligned}$$

$M = \sum_{i=1}^K N_i + \beta$
 $\beta, \Delta_i \in \mathbb{R}; N_i - \text{i.i.d., Gaus.}$
 $\Rightarrow M - \text{Gaus}$

$$\left. \begin{aligned} Z_j &- \text{Gaussian random variable} \\ E[Z_j] &= 0 \\ \text{cov}(Z_j, Z_k) &= E[Z_j Z_k^*] - E[Z_j] E[Z_k^*] = 0 \end{aligned} \right\} \Rightarrow \{Z_j\}_{j=1}^n \text{ - i.i.d., Gauss, centered}$$

$$d) \hat{Z}(t) = Z(t) - \sum_{j=1}^n Z_j \varphi_j(t) \Rightarrow E[\hat{Z}(t) Z_k^*] = 0 \quad \forall k=1,2,\dots$$

$$\begin{aligned} E[\hat{Z}(t) Z_k^*] &= E[Z(t) Z_k^*] - \sum_{j=1}^n \varphi_j(t) E[Z_j Z_k^*] = \\ &= E[Z(t) Z_k^*] - \varphi_k(t) \cdot 2N_0 = E[Z(t) \int_{-\infty}^{+\infty} Z_k^*(z) \varphi_k(z) dz] - \\ &= \varphi_k(t) \cdot 2N_0 = \int_{-\infty}^{+\infty} E[Z(t) Z_k^*(z)] \varphi_k(z) dz = \varphi_k(t) \cdot 2N_0 \quad \textcircled{=} \end{aligned}$$

$$\begin{aligned} \Delta \int_{-\infty}^{+\infty} E[Z(t) Z_k^*(z)] \varphi_k(z) dz &= \int_{-\infty}^{+\infty} R_Z(t-z) \varphi_k(z) dz = \\ &= (R_Z * \varphi_k)(t) = \mathcal{F}^{-1} \left[\underbrace{S_Z(f)}_{2N_0 \cdot \pi(f)} \cdot \underbrace{\varphi_k(f)}_{\substack{\text{cf. } w; w \\ [-w; w]}} \right] = 2N_0 \varphi_k(t) \end{aligned}$$

$$\textcircled{=} 2N_0 \varphi_k(t) - \varphi_k(t) 2N_0 = 0$$


```
import numpy as np
from scipy.signal import hilbert
```

```
PI = np.pi
```

```
class ComplexEnvelopeModulator:
```

```
    def __init__(self, pass_freq):
        self._pass_freq = pass_freq
```

```
    def get_bandpass_signal(self, x_l):
        f_0 = self._pass_freq

        def x(t):
            x_l_t = x_l(t)
            x_i_t = np.real(x_l_t)
            x_q_t = np.imag(x_l_t)

            return (x_i_t * np.cos(2*PI*f_0*t) + x_q_t * ((-1) * np.sin(2*PI*f_0*t)))

        return x
```

```
    def get_analitic_signal(self, x_l):
        f_0 = self._pass_freq

        def x_a(t):
            x_l_t = x_l(t)
            x_i_t = np.real(x_l_t)
            x_q_t = np.imag(x_l_t)

            s_t = np.sin(2*PI*f_0*t)
            c_t = np.cos(2*PI*f_0*t)

            return (0.5)*(x_i_t*c_t - x_q_t*s_t) + (0.5j)*(x_q_t*c_t + x_i_t*s_t)

        return x_a
```

```
    def get_baseband_signal(self, x):
        f_0 = self._pass_freq

        def x_l(t):
            x_t = x(t)
            H_t = np.imag(hilbert(x_t))*0.5

            s_t = np.sin(2*PI*f_0*t)
            c_t = np.cos(2*PI*f_0*t)

            return (x_t*c_t + H_t*s_t) + (1j) * (x_t*(-1)*s_t + H_t*c_t)

        return x_l
```

```
    def get_signalS_envelope_amp(self, x_l):
        def envelope_amp(t):
            x_l_t = x_l(t)
            x_i_t = np.real(x_l_t)
            x_q_t = np.imag(x_l_t)

            return np.sqrt(x_i_t*x_i_t + x_q_t*x_q_t)

        return envelope_amp
```

```
    def get_signalS_envelope_phase(self, x_l):
        def envelope_phase(t):
            x_l_t = x_l(t)
            x_i_t = np.real(x_l_t)
            x_q_t = np.imag(x_l_t)

            return np.arctan(x_q_t/x_i_t)

        return envelope_phase
```

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.fft import fft, ifft, fftfreq, fftshift

from ComplexEnvelopeModulator import ComplexEnvelopeModulator

PI = np.pi

## Model parameters

pass_freq = 9*1e2

AMP_INFO_SPEC = 7*1e1
INFO_SPEC_BANWIND = 9*1e1

Fs = (pass_freq * 2 + INFO_SPEC_BANWIND + 1e1) * 1e1
dt = 1/Fs
t_start = 1e-4
t_end = 0.2

## Spectrum's func of inform signal
def X_l(f):
    def X_l_uniq(f):
        if ((f < -INFO_SPEC_BANWIND/2) or (INFO_SPEC_BANWIND/2 < f)):
            return 0

        return AMP_INFO_SPEC * (f + INFO_SPEC_BANWIND/2) / INFO_SPEC_BANWIND

    result = np.zeros(np.size(f))

    for i in range(0, np.size(f)):
        result[i] = X_l_uniq(f[i])

    return result

## Time-dimention teoretical functions
A = AMP_INFO_SPEC
w = INFO_SPEC_BANWIND / 2
f_0 = pass_freq

def x_l_teor(t):

    return A / (8*(PI**3)*w*(t*t)) * ((2*PI*w*t + 1j)*np.sin(2*PI*w*t) - (2*PI*1j*w*t)*np.cos(2*PI*w*t))

def x_teor(t):

    return np.real(x_l_teor(t) * np.exp(2*PI*1j*f_0*t))

def x_a_teor(t):

    return 0.5*x_teor(t) + (0.5j)*np.imag(x_l_teor(t)*np.exp(2*PI*1j*f_0*t))

def envelope_amp_teor(t):

    return np.abs(x_l_teor(t))

def envelope_phase_teor(t):

    return np.angle(x_l_teor(t))

## Create modulator object
cem = ComplexEnvelopeModulator(pass_freq)

## Find time-dimention function
x_l = cem.get_baseband_signal(x_teor)
x = cem.get_bandpass_signal(x_l_teor)
x_a = cem.get_analitic_signal(x_l_teor)
envelope_amp = cem.get_signalS_envelope_amp(x_l_teor)
envelope_phase = cem.get_signalS_envelope_phase(x_l_teor)

## Calculate error
t = np.arange(t_start, t_end, dt)

err_x_l = np.abs(x_l(t) - x_l_teor(t))
err_x = np.abs(x(t) - x_teor(t))
err_x_a = np.abs(x_a(t) - x_a_teor(t))
err_envelope_amp = np.abs(envelope_amp(t) - envelope_amp_teor(t))
err_envelope_phase = np.abs(envelope_phase(t) - envelope_phase_teor(t))

## Find spectrum
N = np.size(t)
T = dt
freq = fftfreq(N,T)
spec_analitic = X_l(freq)

X_l_teor = np.abs(fft(x_l_teor(t))) * (1.0/N)
X_teor = np.abs(fft(x_teor(t))) * (1.0/N)
X_a_teor = np.abs(fft(x_a_teor(t))) * (1.0/N)

X_l_prac = np.abs(fft(x_l(t))) * (1.0/N)
X_prac = np.abs(fft(x(t))) * (1.0/N)
X_a_prac = np.abs(fft(x_a(t))) * (1.0/N)

## Plotting errors
plt.figure(figsize=(10, 6))

plt.semilogy(t, err_x_l,          '.', label='err x_l')
plt.semilogy(t, err_x,           '.', label='err x')
plt.semilogy(t, err_x_a,         '.', label='err x_a')
plt.semilogy(t, err_envelope_amp, '.', label='err envelope amplitude')
plt.semilogy(t, err_envelope_phase, '.', label='err envelope phase')

plt.xlabel('t')
plt.ylabel('Error')
plt.title('Error curves of all functions')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

# Plotting specturms
fig1, axs = plt.subplots(2, 1, figsize=(10, 8))
```

```
axs[0].plot(freq, X_l_teor,          '.', label='X_l teor')
axs[0].plot(freq, X_teor,           '.', label='X teor')
axs[0].plot(freq, X_a_teor,         '.', label='X_a teor')
axs[0].plot(freq, spec_analitic,    '.', label='X_l analitic')
axs[0].set_title('Spectrum curves of teoretical functions')
axs[0].set_xlabel('freq Hz')
axs[0].set_ylabel('Amp')
axs[0].legend()
axs[0].grid(True)

axs[1].plot(freq, X_l_prac,          '.', label='X_l prac')
axs[1].plot(freq, X_prac,           '.', label='X prac')
axs[1].plot(freq, X_a_prac,         '.', label='X_a prac')
axs[1].plot(freq, spec_analitic,    '.', label='X_l analitic')
axs[1].set_title('Spectrum curves of practical functions')
axs[1].set_xlabel('freq Hz')
axs[1].set_ylabel('Amp')
axs[1].legend()
axs[1].grid(True)

plt.tight_layout()
plt.show()

# Plotting time-dimension evnvelope with basepass signal
fig2, axs2 = plt.subplots(3, 2, figsize=(10, 12))

axs2[0][0].plot(t, x_teor(t),          '.', label='x teor')
axs2[0][0].plot(t, envelope_amp_teor(t), '.', label='envelope amp teor')
axs2[0][0].set_title('Signal curves of teoretical functions with envelope')
axs2[0][0].set_xlabel('time sec')
axs2[0][0].set_ylabel('Amp')
axs2[0][0].legend()
axs2[0][0].grid(True)

axs2[0][1].plot(t, envelope_phase_teor(t),          '.', label='envelope phase teor')
axs2[0][1].set_title('Phase curve of teoretical envelope')
axs2[0][1].set_xlabel('time sec')
axs2[0][1].set_ylabel('radian')
axs2[0][1].legend()
axs2[0][1].grid(True)

axs2[1][0].plot(t, x(t),          '.', label='x prac')
axs2[1][0].plot(t, envelope_amp(t), '.', label='envelope amp prac')
axs2[1][0].set_title('Signal curves of practical functions with envelope')
axs2[1][0].set_xlabel('time sec')
axs2[1][0].set_ylabel('Amp')
axs2[1][0].legend()
axs2[1][0].grid(True)

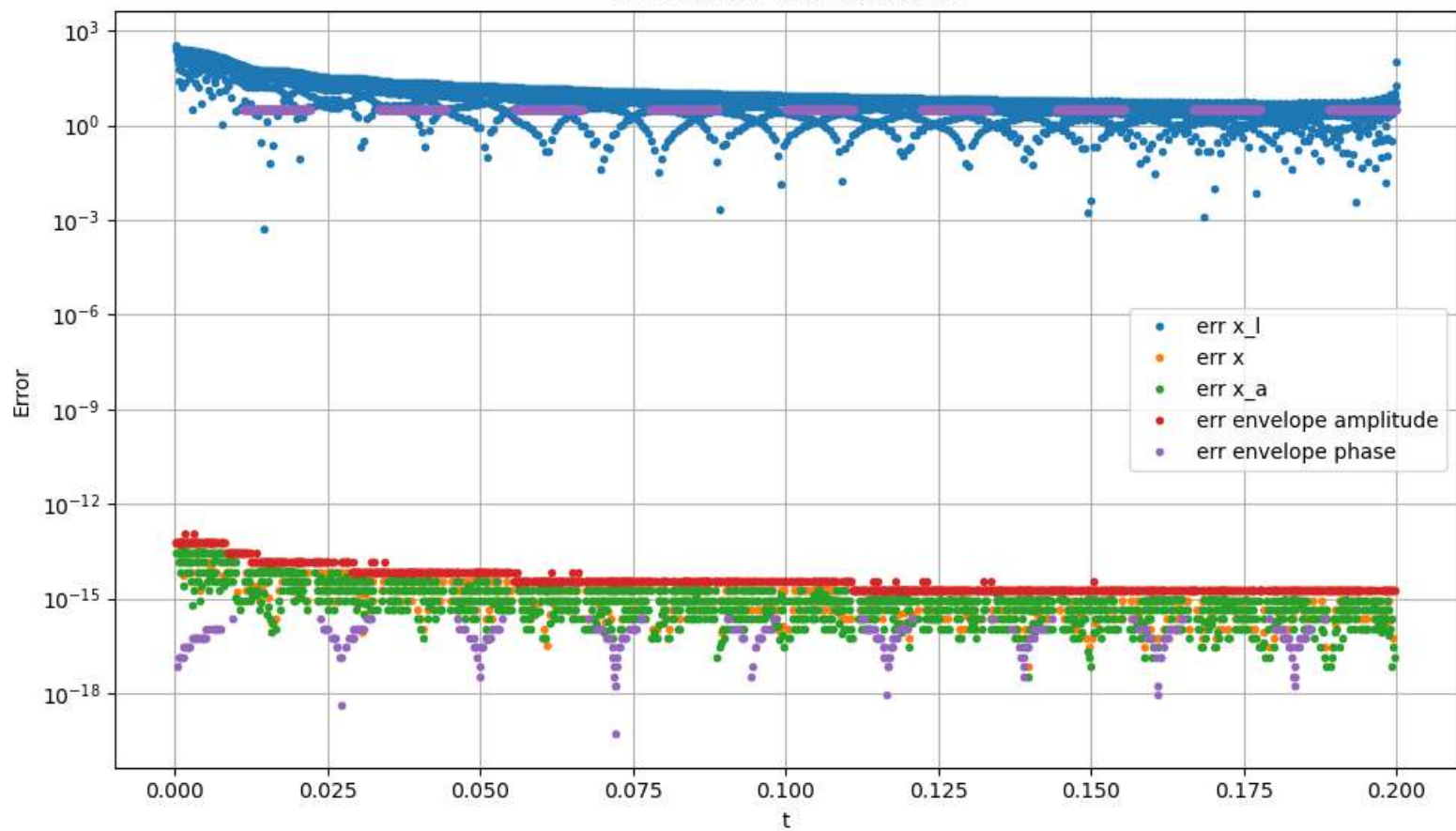
axs2[1][1].plot(t, envelope_phase(t),          '.', label='envelope phase teor')
axs2[1][1].set_title('Phase curve of practical envelope')
axs2[1][1].set_xlabel('time sec')
axs2[1][1].set_ylabel('radian')
axs2[1][1].legend()
axs2[1][1].grid(True)

axs2[2][0].plot(t, x_teor(t),          '.', label='x teor')
axs2[2][0].plot(t, envelope_amp(t),    '.', label='envelope amp prac')
axs2[2][0].set_title('Perform teoretical signal with practical envelope')
axs2[2][0].set_xlabel('time sec')
axs2[2][0].set_ylabel('Amp')
axs2[2][0].legend()
axs2[2][0].grid(True)

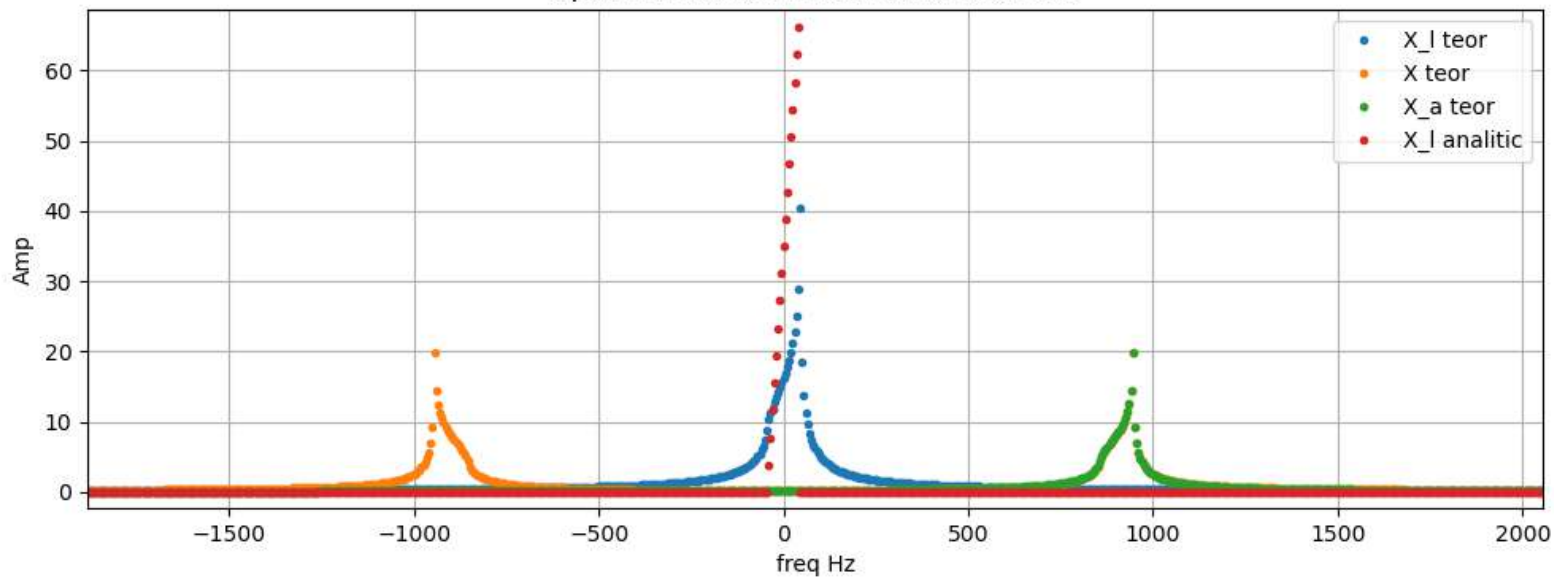
axs2[2][1].plot(t, envelope_phase(t),          '.', label='envelope phase prac')
axs2[2][1].plot(t, envelope_phase_teor(t),    '.', label='envelope phase teor')
axs2[2][1].set_title('Perform phase of envelopes')
axs2[2][1].set_xlabel('time sec')
axs2[2][1].set_ylabel('radian')
axs2[2][1].legend()
axs2[2][1].grid(True)

plt.tight_layout()
plt.show()
```

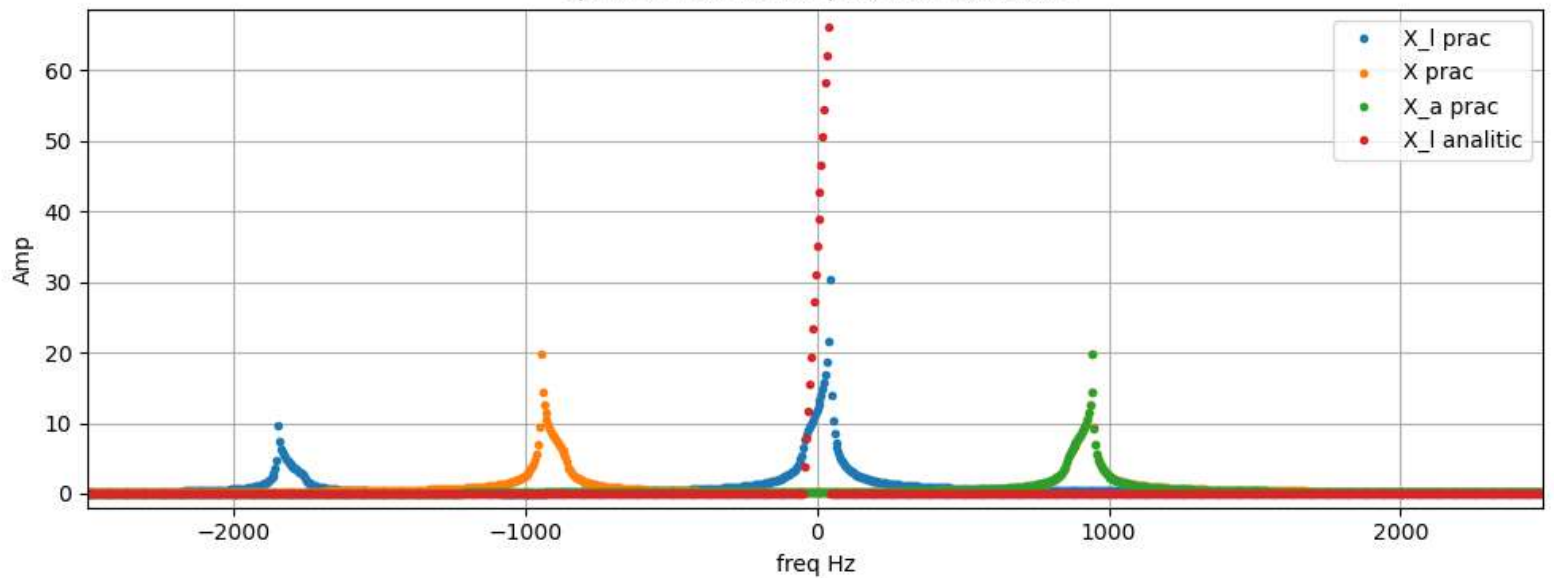
Error curves of all functions



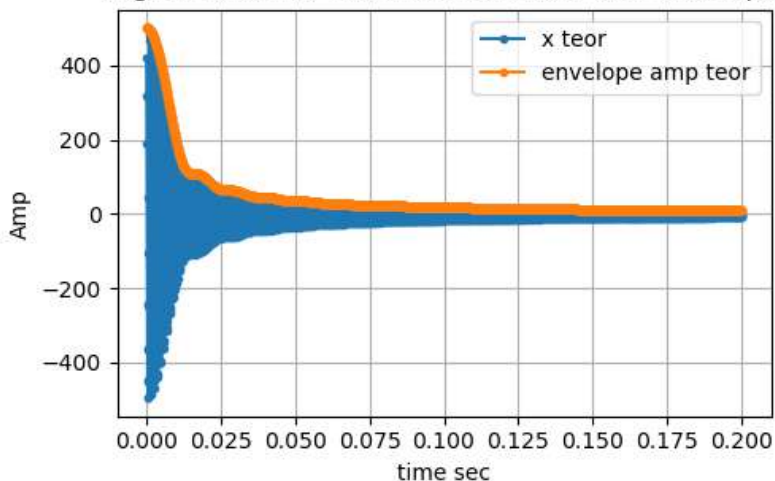
Spectrum curves of teoretical functions



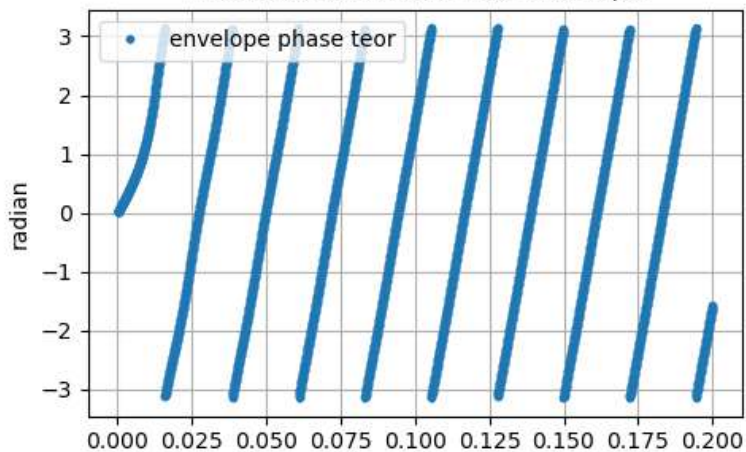
Spectrum curves of practical functions



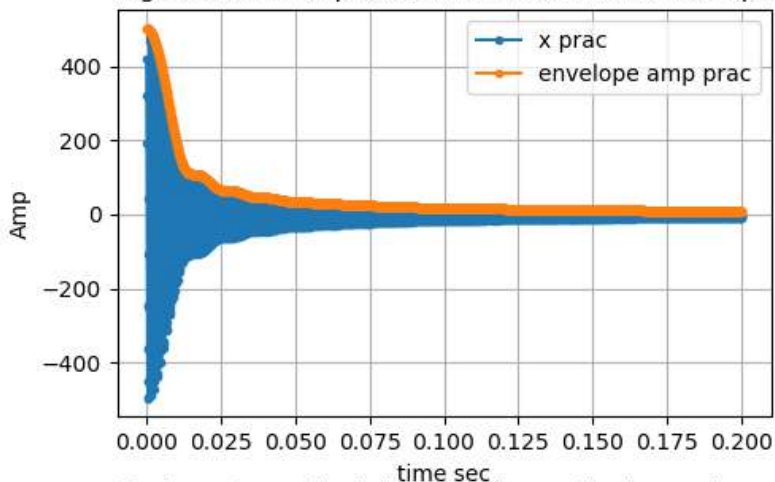
Signal curves of teoretical functions with envelope



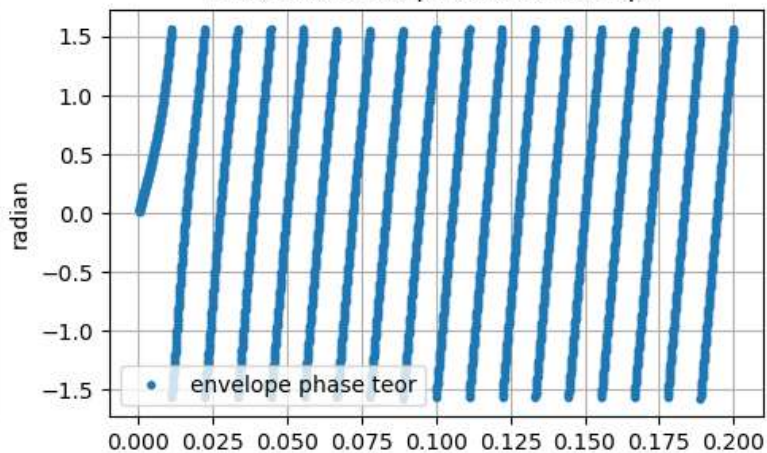
Phase curve of teoretical envelope



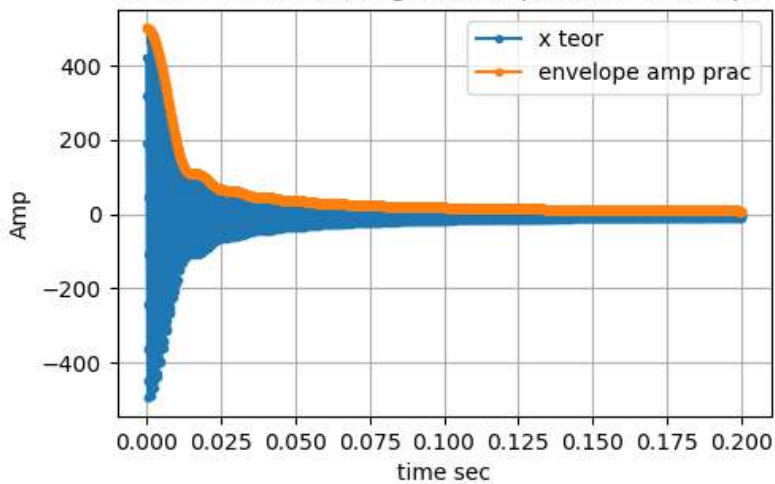
Signal curves of practical functions with envelope



Phase curve of practical envelope



Perform teoretical signal with practical envelope



Perform phase of envelopes

